

## GARRISON, ND, USA

WMO: 727677

Lat: 47.646N

Lon: 101.439W

Elev: 583

StdP: 94.52

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94041

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -28.9      | -26.0 | -32.8                         | 0.2 | -28.7 | -29.5 | 0.3 | -25.8 | 16.3                  | -8.7 | 14.7 | -8.2 | 3.4                   | 300  | 0.680 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.9                   | 32.3            | 20.3 | 30.1 | 19.9 | 28.1 | 18.9 | 22.8                | 28.5 | 21.5 | 27.6 | 20.4 | 26.3 | 5.0                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 21.0                            | 16.8 | 26.2 | 19.5 | 15.3 | 24.8 | 18.3 | 14.2 | 23.3 | 70.4          | 28.1 | 65.3 | 27.5 | 61.3 | 26.4 | 28.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 13.5          | 11.7 | 10.2 | DB                         | -32.9 | 36.2               | 3.2 | 2.2                                                | -35.2 | 37.7       | -37.1 | 39.0       | -38.9 | 40.2       | -41.2 | 41.8 | (4) |
| (5)               |      |      | WB                         | -32.5 | 25.1               | 2.3 | 1.8                                                | -34.2 | 26.4       | -35.6 | 27.5       | -36.9 | 28.5       | -38.6 | 29.8 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 5.1    | -11.3 | -10.9 | -2.9  | 5.0   | 12.0  | 17.3  | 20.4  | 19.7  | 14.8  | 6.7   | -1.8  | -8.9 |
|                                                                               | DBStd       | 12.86  | 8.38  | 7.71  | 7.61  | 5.75  | 4.66  | 3.48  | 3.15  | 3.57  | 4.62  | 5.15  | 6.45  | 7.10 |
|                                                                               | HDD10.0     | 2926   | 661   | 584   | 402   | 166   | 33    | 1     | 0     | 0     | 11    | 128   | 353   | 586  |
|                                                                               | HDD18.3     | 5029   | 919   | 818   | 659   | 401   | 203   | 58    | 14    | 27    | 124   | 360   | 603   | 844  |
|                                                                               | CDD10.0     | 1137   | 0     | 0     | 1     | 16    | 94    | 219   | 324   | 301   | 154   | 27    | 1     | 0    |
|                                                                               | CDD18.3     | 199    | 0     | 0     | 0     | 0     | 5     | 27    | 80    | 69    | 17    | 1     | 0     | 0    |
|                                                                               | CDH23.3     | 2310   | 0     | 0     | 0     | 12    | 96    | 299   | 837   | 783   | 264   | 19    | 0     | 0    |
|                                                                               | CDH26.7     | 763    | 0     | 0     | 0     | 2     | 21    | 71    | 282   | 301   | 83    | 3     | 0     | 0    |
| Wind                                                                          | WSAvg       | 4.5    | 4.7   | 4.5   | 4.8   | 4.9   | 4.9   | 4.4   | 3.6   | 3.8   | 4.1   | 4.8   | 4.8   | 4.6  |
| Precipitation                                                                 | PrecAvg     | 475    | 12    | 10    | 17    | 34    | 78    | 107   | 70    | 54    | 37    | 30    | 15    | 13   |
|                                                                               | PrecMax     | 689    | 27    | 61    | 35    | 74    | 178   | 274   | 221   | 190   | 86    | 170   | 54    | 45   |
|                                                                               | PrecMin     | 263    | 0     | 0     | 2     | 6     | 20    | 53    | 29    | 12    | 11    | 2     | 0     | 1    |
|                                                                               | PrecStd     | 94     | 8     | 12    | 10    | 18    | 38    | 44    | 39    | 43    | 21    | 34    | 14    | 10   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.3   | 7.4   | 19.2  | 24.7  | 29.3  | 31.7  | 34.7  | 35.3  | 32.8  | 26.0  | 17.0  | 8.1  |
|                                                                               |             | MCWB   | 3.4   | 3.5   | 9.7   | 12.7  | 16.3  | 20.4  | 21.3  | 20.6  | 18.7  | 14.8  | 9.6   | 3.9  |
|                                                                               | 2%          | DB     | 4.0   | 4.0   | 14.1  | 20.9  | 26.0  | 28.5  | 31.9  | 32.5  | 29.0  | 21.6  | 12.4  | 4.5  |
|                                                                               |             | MCWB   | 1.5   | 1.5   | 7.9   | 10.6  | 15.3  | 19.2  | 21.1  | 20.0  | 17.4  | 13.1  | 6.3   | 1.7  |
|                                                                               | 5%          | DB     | 2.1   | 1.9   | 10.1  | 17.8  | 23.4  | 26.6  | 29.8  | 30.1  | 26.6  | 18.0  | 9.5   | 2.5  |
|                                                                               |             | MCWB   | 0.4   | 0.1   | 5.8   | 9.2   | 14.0  | 18.2  | 21.2  | 19.4  | 16.5  | 10.7  | 5.2   | 0.4  |
|                                                                               | 10%         | DB     | 0.1   | 0.1   | 6.3   | 15.0  | 21.0  | 24.8  | 27.8  | 27.7  | 23.8  | 15.0  | 6.7   | 0.7  |
|                                                                               |             | MCWB   | -1.1  | -1.2  | 3.5   | 7.8   | 12.8  | 17.1  | 19.9  | 18.6  | 15.4  | 9.5   | 3.4   | -0.8 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.6   | 3.9   | 10.6  | 13.9  | 19.0  | 23.4  | 25.4  | 23.4  | 21.0  | 16.4  | 10.1  | 4.1  |
|                                                                               |             | MCDB   | 7.0   | 6.9   | 17.6  | 23.1  | 25.0  | 28.3  | 30.7  | 30.1  | 28.3  | 24.0  | 16.2  | 7.6  |
|                                                                               | 2%          | WB     | 1.7   | 1.7   | 8.0   | 11.3  | 16.9  | 21.2  | 23.5  | 21.8  | 19.4  | 13.6  | 7.2   | 1.9  |
|                                                                               |             | MCDB   | 3.7   | 3.9   | 13.9  | 19.0  | 22.9  | 26.6  | 28.8  | 29.1  | 25.9  | 19.5  | 11.6  | 4.2  |
|                                                                               | 5%          | WB     | 0.4   | 0.3   | 5.7   | 9.9   | 15.4  | 19.7  | 22.1  | 20.7  | 17.8  | 11.9  | 5.3   | 0.5  |
|                                                                               |             | MCDB   | 2.1   | 1.7   | 10.0  | 16.7  | 21.0  | 24.4  | 27.5  | 27.9  | 23.9  | 17.3  | 8.9   | 2.3  |
|                                                                               | 10%         | WB     | -1.1  | -1.2  | 3.6   | 8.3   | 13.9  | 18.2  | 20.9  | 19.8  | 16.3  | 10.1  | 3.7   | -0.9 |
|                                                                               |             | MCDB   | 0.2   | 0.0   | 6.3   | 14.2  | 19.3  | 23.2  | 26.6  | 25.7  | 22.6  | 14.6  | 6.5   | 0.6  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.8   | 10.2  | 10.5  | 12.5  | 12.6  | 11.7  | 12.9  | 13.9  | 13.4  | 11.2  | 10.0  | 8.9  |
|                                                                               |             | MCDBR  | 10.8  | 11.4  | 15.9  | 17.9  | 16.7  | 15.0  | 15.7  | 17.3  | 18.0  | 16.9  | 14.2  | 11.2 |
|                                                                               |             | MCWBR  | 8.9   | 9.2   | 10.2  | 9.6   | 8.0   | 7.8   | 7.4   | 7.3   | 8.3   | 9.4   | 9.2   | 8.9  |
|                                                                               | 5% WB       | MCDBR  | 10.4  | 10.9  | 15.0  | 17.0  | 13.3  | 12.7  | 13.4  | 14.6  | 15.4  | 15.6  | 13.6  | 10.9 |
|                                                                               |             | MCWBR  | 8.7   | 9.0   | 10.0  | 9.6   | 7.1   | 7.8   | 7.4   | 6.9   | 7.7   | 9.1   | 9.1   | 8.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.252  | 0.259 | 0.286 | 0.324 | 0.344 | 0.366 | 0.374 | 0.381 | 0.336 | 0.302 | 0.267 | 0.247 |      |
|                                                                               | taud        | 2.435  | 2.451 | 2.483 | 2.412 | 2.417 | 2.391 | 2.396 | 2.351 | 2.453 | 2.526 | 2.501 | 2.460 |      |
|                                                                               | Ebn at Noon | 853    | 926   | 946   | 930   | 918   | 896   | 883   | 860   | 874   | 860   | 828   | 818   |      |
|                                                                               | Edh at Noon | 64     | 79    | 90    | 108   | 111   | 115   | 113   | 114   | 93    | 74    | 60    | 55    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1.62   | 2.70  | 3.86  | 4.86  | 5.74  | 6.28  | 6.64  | 5.50  | 4.11  | 2.57  | 1.57  | 1.15  |      |
|                                                                               | RadStd      | 0.16   | 0.28  | 0.31  | 0.34  | 0.45  | 0.47  | 0.28  | 0.36  | 0.28  | 0.30  | 0.23  | 0.11  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (11 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page